Mobin Shahidi, M.Sc.

Full-stack AI Engineer | Production Systems

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PROFESSIONAL SUMMARY

AI Engineer with **3+ years** of experience building **ML** systems, **NLP** applications, **Recommender Systems**, and **data pipelines**, combined with **4+ years** of **full-stack web development** experience. Proven track record of designing and deploying production ML systems using **AWS** with **CI/CD** and infrastructure as code. Strong in **Python**, **C#, FastAPI**, and system design, with hands-on experience building scalable, real-time **LLM** applications end-to-end.

KEY SKILLS

AI / ML
- ✓ Recommender Systems
- ✓ Production ML
- ✓ XAI, NLP, Data Pipeline
- ✓ LLM (Fine-tuning, RAG)

Engineering
- ✓ Performance Evaluation
- ✓ System Architecture
- ✓ REST API Design
- ✓ System Monitoring

Backend / Frontend
- ✓ Python, C#
- ✓ FastAPI, TensorFlow, PyTorch
- ✓ Next.js, TypeScript, JavaScript
- ✓ SQL + Databases

Infrastructure
- ✓ Docker + ECR + CloudWatch
- ✓ Terraform (IaC)
- ✓ CI/CD (GitHub Actions, Vercel)
- ✓ AWS (SageMaker, Lambda)

WORK EXPERIENCE

AI Research Engineer, University of Potsdam, Full-time *Feb 2024 – Feb 2026*
Project: Recommender Systems for Quality Political News Source *Potsdam, Germany*

- Built an end-to-end **data pipeline in Python** (Pandas, NumPy) converting +5M tweets into **41 engineered information-quality signals** for ML modeling and recommendation tasks.
- Implemented reproducible preprocessing and feature extraction workflows with **data validation checks and PyTest-based tests** to ensure dataset consistency and experiment repeatability.
- Trained supervised ML models using **Scikit-learn** for news source quality prediction using **structured training pipelines**, cross-validation, and hyperparameter tuning, achieving 0.58 correlation against NewsGuard benchmark ratings.
- Applied **model versioning and experiment tracking** practices to compare runs and maintain reproducible ML experiments.
- Dockerized ML and recommendation models and deployed them on AWS, implementing performance logging and basic observability to monitor model behavior and pipeline execution.
- Designed a **quality-aware recommender system** that increased the average quality score of recommended news sources by **+3 points** across users while keeping precision unchanged versus baseline.
- Collaborated in a **cross-functional team** and presented experiment results and model performance insights to stakeholders.

Research Assistant, Bu-Ali Sina University, Full-time *Mar 2023 – Sep 2023*
Project: Explainable Recommendation System *Hamedan, Iran*

- Designed a large-scale **NLP + recommendation framework** over **151K reviews/ratings (14K users, 8.7K items)** using multi-task learning to jointly model user preferences and generate explainable recommendations.
- Built **text generation and processing pipelines** that produce personalized explanations by combining neural generation models with summarization, sentiment filtering, user-similarity selection, and style-conditioned rewriting.
- Implemented and evaluated neural text generation and personalization models in Python/TensorFlow, achieving **~0.97 semantic similarity** in generated explanations while improving recommendation prediction performance (**MSE 0.992**).

Full-stack Web Developer, Computer Afzar Hamedan (CAH), Co-founder *Feb 2017 - Jul 2021*
Start-up Project: Cross-Platform Image Management and Privacy Solution (PhotoCans) *Hamedan, Iran*

- Delivered the full-stack web platform for the commercial launch of a cross-platform image management product serving **100+ users**, using ASP.NET Core, **C#**, JavaScript, CSS, and relational databases across product, customer, and sales modules.
- Implemented **33 REST API endpoints** and backend services for product operations, licensing, and system integration, improving responsiveness and throughput through asynchronous and non-blocking request handling.
- Designed modular system architecture and database schemas and implemented **12+ business workflows**, adding testing practices that improved application robustness and maintainability.
- Deployed and maintained the production system on IIS infrastructure across multiple releases, resolving operational issues through logging, diagnostics, and structured troubleshooting.

EDUCATION

Master **Artificial Intelligence and Robotics,** Bu-Ali Sina University *Jan 2020 - Jan 2023*

Bachelor **Software Engineering,** Bu-Ali Sina University *Sep 2014 - Jul 2019*

LANGUAGES

Persian: Native **English**: Advanced (C1) **German**: Conversational (B1)

SIDE PROJECTS

News Source Quality Platform (www.meydanews.com) *Mar 2026 – Present*

- Developed a **ML model** to **score news source quality** using engineered network, structural, and diversity features from X (Twitter) data, deployed via **AWS SageMaker** real-time endpoint.
- Architected a serverless backend using **FastAPI on AWS Lambda** (container-based) with API Gateway, enabling scalable, low-latency inference.
- Containerized the backend with Docker and integrated SageMaker inference with **S3-based** feature retrieval and **caching** for performance optimization.
- Built a **Next.js (TypeScript)** frontend deployed on Vercel with environment-based API integration for real-time predictions.
- Implemented Infrastructure as Code (**Terraform**) and CI/CD pipelines (GitHub Actions, Vercel) to automate deployment and manage AWS resources (Lambda, API Gateway, IAM, **ECR**), with monitoring via **CloudWatch**.

Chatbot Evaluation Framework (Link) *Feb 2026 – Feb 2026*

- Developed a Python-based **chatbot** evaluation toolkit to benchmark **LLM responses** across multiple prompts.
- Implemented automated evaluation pipelines comparing generated answers with reference responses.
- Created tools to measure response quality, consistency, and correctness in **conversational AI** systems.
- Built reproducible testing workflows for systematic validation of chatbot performance.

Translation Quality Evaluation (Link) *Jan 2024 - Jan 2024*

- Built an **automatic translation pipeline** to process datasets end-to-end, including preprocessing, translation, and post-processing steps.
- Experimented with **fine-tuning translation models** on domain-specific data to improve translation quality and consistency.
- Implemented an **evaluation workflow** to compare translation outputs using basic quality metrics and reference-based checks.

Workflow-Aware AI Assistant for Applications *Nov 2025 – Jan 2026*

- Built a workflow-aware AI assistant integrated into a **Next.js** application, responding based on application state and user action order.
- Implemented backend orchestration, integrating both **OpenAI APIs** and **local LLMs (Mistral)** for controlled and flexible inference, using **LangChain**.

Deep Learning NLP Recommender System with Reinforcement Learning (Masters' Thesis) *Oct 2021 - Jan 2023*

- Built a deep learning explainable recommender system in **Python (TensorFlow, PyTorch)** trained on 151K reviews/ratings, combining NLP and multi-task learning for personalized recommendations and explanations.
- Implemented neural text modeling pipelines using **RNN, CNN, and transformer-based encoders** for review understanding and explanation generation.
- Applied **reinforcement-learning**–inspired and neural text generation methods to optimize explanation selection and reconstruction, achieving MSE 0.992 in rating prediction.
- Optimized end-to-end training and inference pipelines and executed large-scale model training on **GPUs,** significantly reducing training time and improving experimentation throughput.

PUBLICATIONS

- *Exogenous Cues to Information Quality (2026)*
- *Improving News Reliability in Algorithmic Newsfeeds (2026)*
- *Possible Futures: Societal Consequences of Algorithmic Design and Social Media Use (2026)*
- *A Framework for Accurate Recommendations and Explanation Generation Using Multi-Task Learning (2023)*